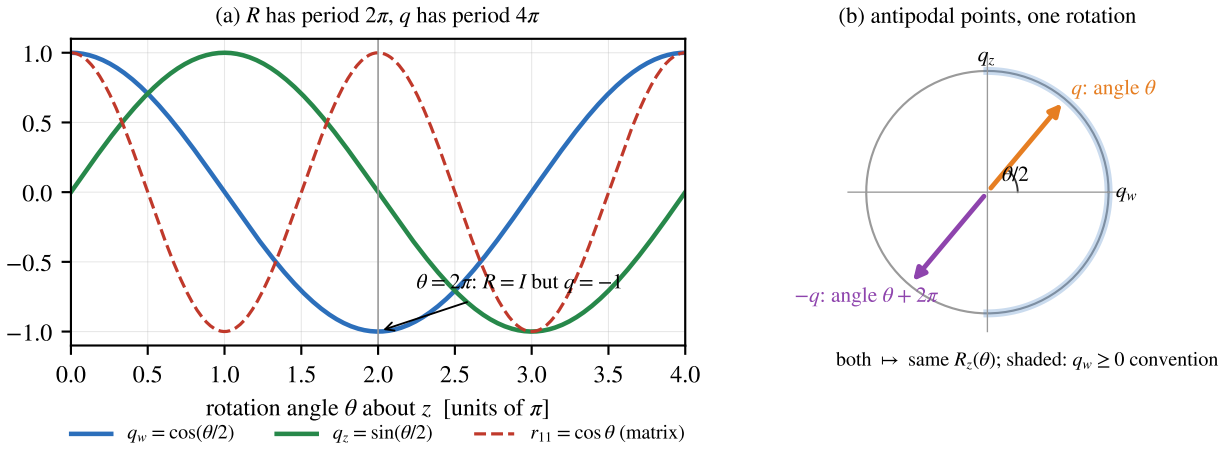
(a) R has period 2π, q has period 4π
1.0
0.5
0.0
−0.5
−1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
$\theta = 2\pi$: $R = I$ but $q = -1$
rotation angle θ about z [units of π]
$q_w = \cos(\theta/2)$
$q_z = \sin(\theta/2)$
$r_{11} = \cos\theta$ (matrix)
(b) antipodal points, one rotation
q_z
q: angle θ
$\theta/2$
q_w
$-q$: angle $\theta + 2\pi$
both ↦ same $R_z(\theta)$; shaded: $q_w \geq 0$ convention